

(අස්වැන්න)

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:	() — 2025 / 2026
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:	(, , , , , , ,)

15 — :

#			
1	...	230544	, (/), ,
2	... ()	230476	, 4- -
3	...	230487	, ' , ,
4	... ()	230503	(), ,
5	...	230467	2 - 5 ,

: - (,
) , ,
, , - .

1. - (-)

' (, , ,),
30% 40% (. 180).
' ' (වගා රුල්ල).

(ගොවි මහතුන්ට සහ සාමාන්‍ය පරිශීලකයන්ට):

'ගොවි මහත්තුරු වළඳපොළේ අද තියනේ මල බලලා හැමෝම එකම බෝගය (උදා: ලීක්ස්, ගෝවා) එකවර වගා කරනවා. මාස 3කට පස්සේ හැමෝගමෙ අස්වැන්න එකම සතියේ වළඳපොළට ආවම මල උපියල් 20-30ට බහිනවා. මේකට තමයි වගා රුල්ල කියන්නේ. කියන්නේ මේ ප්‍රශ්නය මුලින්ම විසඳන ඩිජිටල් පද්ධතියක්.'

'1. ගොවියා බිජ දන්න කලින් ඇප් එකෙන් බලනවා තමන්ගේ කලාපයේ ඒ බෝගය දැනටමත් කොඵ්වර වගා කරලා තියනෙවද කියලා.

2. වගාව 85% ඉක්මවා ඇත්නම් (-), පද්ධතිය රතු එළියක් දල්වා වැඩි ලාභයක් ලබාගත හැකි වනෙත් බෝග (බේට්ටර්, නේකර්ල්) නිර්දේශ කරයි.

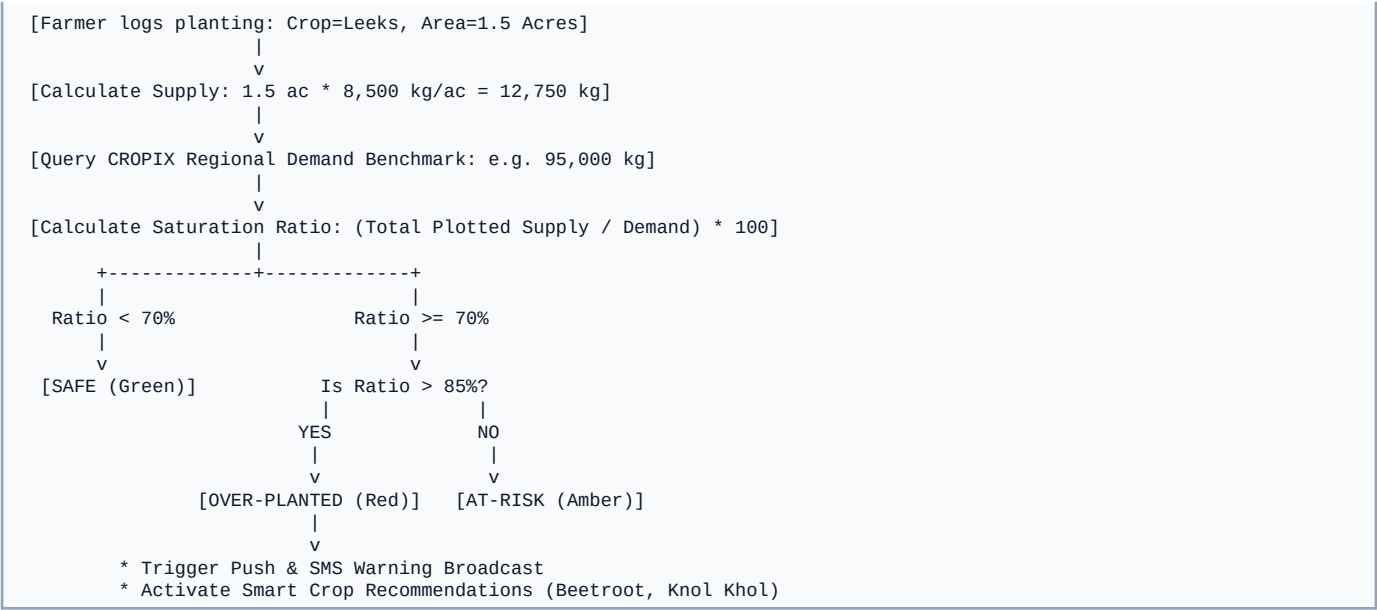
3. ස්මාර්ට්ෆෝන් නැති ගොවි මහතුන්ගේ දත්ත ගොවිජන සේවා නිලධාරියා () විසින් වඩේ පද්ධතියට ඇතුළත් කරයි ().

4. ඉතිරි වන අතිරික්ත අස්වැන්න කිලෝමීටර් 5ක් ඇතුළත හෝටල්, වළඳුන්ට කළේන්ම චිකුණා ගැනීමට සෘජු වළඳපොළක් සපයයි.'

2.

(-)

PRESENTATION TIER (CLIENTS)			
Farmer & Buyer Mobile App (Flutter)		Divisional Officer Portal (React)	
- GPS Planting Logger		- Cultivation Map & Saturation	
- Over-Planting Push Alerts (FCM)		- Officer Proxy Data Entry Form	
- Offline Queue Storage		- Multi-Channel Warning Broadcast	
- 5 km Surplus Produce Feed		- Trilingual UI (EN / SI / TA)	
REST API (HTTP/JSON)			
APPLICATION & LOGIC TIER			
Node.js / Express.js REST API Server (Port 5000)			
PREDICTIVE RISK ENGINE		Supply Density vs CROPIX Demand Analyzer	
SMART RECOMMENDATION ENGINE		4-Factor Composite Agro-Scoring Matrix	
GEOFENCING SERVICE		Haversine 5 km Radius Matching Engine	
NOTIFICATION SERVICE		Firebase Cloud Messaging + SMS Fallback	
Relational SQL		Realtime Sync	
DATA PERSISTENCE TIER			
PostgreSQL 15 Database (Port 5432)		Firebase Realtime Database (Cloud)	
- 10 Relational Tables & Indexes		- Live produce listing sync	



3. 15 — 5 '

1: . . . (230544)

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:
///. 10 ( , ,
_ _ _ _ _ ).
///. -
(24 , 8 ).
///. ,
, 3- .
```

Mathematical Formula:

```
Estimated_Supply = SUM(Planted_Acres * Avg_Yield_Per_Acre_Kg)
Risk_Percentage = (Estimated_Supply / CROPIX_Regional_Quota_Kg) * 100
• Safe: < 70% | Warning: 70% to 85% | Over-Planted: > 85%
:
```

```
cd backend && npm run migrate && npm run seed && curl -X GET http://localhost:5000/api/v1/risk/crop/1
```

2: . . . (- 230476)

```
: - - .
:
///.
///..
-,
4 :
1. (35% ) — .
2. (25% ) — / .
3. (20% ) — 14-22 .
4. (20% ) — 3 .
5 // .
```

Formula: Composite = (0.35 * MarketGap) + (0.25 * Soil) + (0.20 * Weather) + (0.20 * Price)

```
:
```

```
curl -X GET "http://localhost:5000/api/v1/recommendations?district=Badulla&cropId;=1"
```

3: . . . (230487)

```
: () ,
.
:
. (///, ..
.).
///.
.
///.
///.
0 .
/// ( , ).
: ://:3000, 0771234567 / 123, '+
' .
```

4: (- 230503)

: - ,
.
:
 (//, _,"
_).
 ///_.
 .
 ///_.
 , - .
 //10/ , , .
:
 , , , ,
 .

5: . . . (230467)

: 2 - - , 5
.
:
 ///. ///.
-
 ///.
 .
 ///. 30-
- - .
 ///.
 ///.

Evaluator Verification: Query GET /api/v1/marketplace/search-nearby?lat=6.8258&lng;=80.9982&radius;_km=5.

4. --

, , , :

//. —	Test: node src/server.js
:	, , .
:	app.js, config.js
:	PORT=5000

//. —	Test: npm start
:	(,), , (), 1 .
:	express, cors, helmet, routes
:	None

/// . —	Test: node -e 'require("./src/config/config")'
:	(.) , , , .
:	dotenv
:	DB_*, JWT_*, RISK_*

/// . —	Test: npm run migrate
:	". .
:	pg, config.js
:	DB_HOST, DB_NAME

/// . —	Test: Trigger notifications
:	.
:	firebase-admin
:	FIREBASE_PRIVATE_KEY

/// . —	Test: psql -d asvanna_db -f schema.sql
:	10 : , , , _ , , , , .

:	PostgreSQL 12+
:	asvanna_db
/// .-	Test: npm run migrate
:	. , , .
:	fs, database.js
:	DB credentials
/// .-	Test: npm run seed
:	8 , 5 (, , ,) , .
:	bcryptjs, database.js
:	DB credentials
// /. -	Test: curl with Bearer Token
:	- (, , ,).
:	jsonwebtoken
:	JWT_SECRET
// /. -	Test: Trigger 404 / 500 error
:	.
:	apiResponse.js
:	NODE_ENV
// / .-	Test: Send invalid payload
:	/ - 400 .
:	express-validator
:	None
/// .-	Test: node test/test_all_endpoints.js
:	- () .
:	Math library

:	None
/// .-	Test: Imported in controllers
:	: { , , , }.
:	None
:	None
/// .-	Test: GET /api/v1/risk/crop/1
:	' ' 3- .
:	database.js, config.js
:	RISK_SAFE_THRESHOLD
/// .-	Test: GET /api/v1/recommendations
:	4- (, , ,).
:	riskEngineService.js
:	None
/// .-	Test: GET /marketplace/search-nearby
:	5 - 20 .
:	haversine.js
:	DEFAULT_GEOFENCE_RADIUS_KM
/// .-	Test: POST /api/v1/broadcasts
:	.
:	firebase.js, database.js
:	SMS_GATEWAY_URL
/// .-	Test: Publish surplus listing
:	.

:	firebase.js
:	FIREBASE_DATABASE_URL
// /. —	Test: POST /api/v1/auth/login
:	‘ ‘ ‘ ‘ .
:	bcryptjs, jsonwebtoken
:	JWT_SECRET
// /. —	Test: POST /api/v1/planting/log
:	‘ ‘ .
:	database.js, riskEngine
:	None
// /. —	Test: GET /api/v1/risk/regional-summary
:	- .
:	riskEngineService.js
:	None
// /. —	Test: GET /api/v1/recommendations
:	.
:	recommendationService.js
:	None
// /. —	Test: POST /api/v1/marketplace/orders
:	‘ ‘ .
:	geofencingService.js
:	None

// /. —	Test: GET /api/v1/officer/farmers
:	- .
:	database.js, bcryptjs
:	None
// /. —	Test: POST /api/v1/broadcasts
:	- .
:	notificationService.js
:	None
//. —	Test: npm run dev
:	- , , .
:	react-router-dom, contexts
:	None
/// . —	Test: Login / Logout actions
:	, , .
:	api.js
:	asvanna_token
/// . —	Test: Navbar language toggle
:	(, ,) .
:	locales/*.json
:	asvanna_lang
/// . —	Test: View http://localhost:3000
:	, , .
:	StatCard, Modals
:	None

/// .-	Test: View /monitoring
:	''
:	api.js
:	None
/// .-	Test: View /risk-analytics
:	4-
:	api.js
:	None
/// .-	Test: View /farmers
:	.
:	api.js
:	None
/// .-	Test: View /broadcasts
:	.
:	BroadcastModal.jsx
:	None
/// .-	Test: View /marketplace
:	5 .
:	api.js
:	None
// /. -	Test: Click '+ Proxy Data Entry'
:	.
:	api.js
:	None

// /_ —	Test: Click 'Broadcast Warnings'
:	.
:	api.js
:	None
//. —	Test: flutter run
:	, , .
:	flutter_localizations
:	None
/// /_ —	Test: Trigger on mobile
:	.
:	geolocator
:	GPS Permission
/// /_ . —	Test: Test in Airplane mode
:	.
:	shared_preferences
:	None
/// /_. —	Test: Bottom nav switcher
:	, -, .
:	planting, recs, market
:	None
/// /_ . —	Test: Tap 'Log Planting'
:	.
:	location_service.dart
:	None

/// /_ — . —	Test: Tap 'Recommendations' tab
:	.
:	app_colors.dart
:	None
/// /_ ._ —	Test: Tap 'Marketplace' tab
:	5 .
:	list_surplus_screen.dart
:	None
._ —	Test: docker-compose up --build -d
:	- 15, . , .
:	Docker Engine
:	POSTGRES_DB, PORT
./. —	Test: git push origin main
:	‘ ‘ .
:	GitHub Actions
:	NODE_ENV=test

5.

:

1:

:	node backend/test/test_all_endpoints.js
:	4 , 3- , . 4/4 .

2:

:	cd backend && npm run migrate && npm run seed
:	(10) 8 , 5 , .

3:

:	cd backend && npm run dev
:	://:5000. ://:5000/ ({ : , : " }).

4:

:	cd frontend && npm run dev
:	://:3000. (0771234567 / 123) .

5:

:	cd mobile && flutter pub get && flutter run
:	, , .

6: -

:	docker-compose up --build -d
:	- (, ,) .